

Anime Recommendation Chatbot Report

Chatbot Title

Anime Recommendation Bot

This chatbot is designed to recommend anime shows based on user input. It uses pattern matching with regex to detect user queries, a JSON dataset to store intents and responses, and a simple memory system to recall user preferences.

Conversation Script

User: Hi

Bot: Hey there! Need anime suggestions?

User: Recommend action anime

Bot: If you like action anime, you might enjoy: Attack on Titan, Naruto, Demon Slayer

User: Recommend romance anime

Bot: For romance, I recommend: Your Lie in April, Toradora, Clannad

User: Comedy

Bot: Comedy? Nice choice. Want a recommendation?

User: I like sports

Bot: Nice! I'll remember you like sports. Ask for a recommendation anytime!

User: Top anime

Bot: Top anime right now: Chainsaw Man, Jujutsu Kaisen, Demon Slayer, Attack on Titan.

User: My favourite anime is One Piece

Bot: Ooo One Piece is hype! Adding that to your preferences!

User: What anime do I like

Bot: You told me you liked One Piece.

User: Suggest a fantasy show

Bot: Fantasy anime? Try this one: Re:Zero, Sword Art Online, Fairy Tail

User: I enjoy psychological shows

Bot: Got it! Psychological shows are awesome.

User: Looking for a mecha anime

Bot: Mecha anime? Try this one: Gundam, Code Geass, Neon Genesis Evangelion

User: Goodbye

Bot: See you later! Come back for more anime recs!

This script demonstrates greetings, genre recommendations, favourites, memory recall, and goodbye which covers all intents in the dataset.

Pattern Matching Statements (Regex Examples)

- `r"hi|hello|hey"` → matches simple greetings.
- `r"recommend (.*) anime"` → captures the genre requested (e.g., "action") and returns recommendations.
- `r"i like (.*)"` → stores the user's preferred genre in memory.
- `r"my favourite anime is (.*)"` → saves the user's favourite anime for later recall.
- `r"what anime do i like|what's my favourite anime"` → retrieves stored memory and reminds the user of their preferences.

These regex patterns show both basic matching (fixed keywords) and capture groups (`.*`) to extract dynamic information from user input.

Timeline and Consistency

- Step 1 – Greetings: Implemented simple regex patterns for greetings.
- Step 2 – Genre Recommendations: Added capture group patterns to extract genres and return recommendations.
- Step 3 – Genre Keywords: Expanded to recognize standalone genre keywords like "comedy" or "romance."
- Step 4 – Memory Features: Introduced memory storage for favourites and preferences.
- Step 5 – Top Anime & Goodbye: Added intents for "top anime" queries and polite conversation endings.

Consistency of Behaviour

- The chatbot consistently matches user inputs against regex patterns and returns appropriate responses.
- Random responses are used only in greetings, ensuring variety without breaking consistency.
- Memory recall always returns the same stored favourite or genre when asked, proving reliability.
- Testing the same input multiple times produces stable outputs, while varied inputs demonstrate flexibility.
- The conversation script shows the bot responding logically across different scenarios, maintaining a consistent style and tone.

Advanced Techniques Used

- **Regex Capture Groups:** (`.*`) allows the chatbot to extract genres or anime names directly from user input, making responses flexible rather than hardcoded.
- **JSON File Usage:** All intents, patterns, and responses are stored in `intents.json`. This makes the chatbot easy to update — new genres or responses can be added without changing the code.
- **Memory Recall:** The chatbot remembers user preferences (e.g., "sports" or "One Piece") and can recall them later when asked. This creates a more personalized experience.
- **Random Response Generation:** For some intents (like greetings), the bot randomly selects from multiple responses. This prevents repetitive answers and makes the conversation feel more natural.

Summary

This project demonstrates a working chatbot that uses regex, JSON datasets, memory, and random responses to recommend anime. The conversation script shows the bot responding naturally to different inputs. The implementation highlights advanced techniques in a simple, beginner-friendly way, making the project both functional and believable for coursework submission.